Americas Benchmarking Software

Focus shifts to profitable growth

In this report, we update our quarterly dataset of KPIs for Software companies in our coverage. Key observations:

- 4Q21 Software fundamentals were solid with the median Software company beating Street revenue expectations by 3.0% (vs. 3.6% in 3Q21 and 4.3% in 4Q20, FactSet). By segment, fundamentals were particularly robust in Workflow Automation and Security, and more muted in uCaaS, Productivity, and eCommerce.
- However, the median LTV:CAC declined to 4.06 vs. 4.17 in 3Q21, driven by an increase in CAC after 5 quarters of declines. A key bifurcation on how stocks traded on EPS reports was guidance for operating leverage in 2022: companies guiding to margin expansion or better-than-anticipated FCF were generally received positively; in contrast, companies guiding to yoy margin declines or a change in trajectory to breakeven FCF saw outsized underperformance in the following session.
- The changing rates backdrop has increased investor focus on strong unit economics (vs. revenue growth at any cost), with EV/sales multiple correlations with LTV:CAC and the Rule of 40 increasing, and correlations with revenue growth decreasing.

We highlight the following Buy rated stocks where we expect to see a balance between steady revenue growth and expanding EBIT margins over the medium term: ADBE, CRM (on CL), HUBS, NOW (on CL), RPD, TENB, RNG, WDAY, and XM (on CL).

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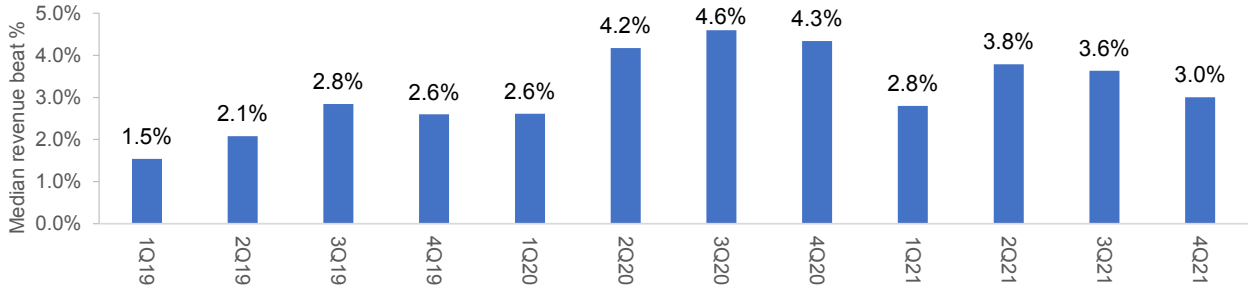
4Q21 Benchmarking Highlights

1) Software fundamentals remain robust, albeit with post-COVID moderation

The median Software company in our sample set beat 4Q21 revenue estimates (FactSet) by 3.0%, compared to 3.6% in 3Q21 and 4.3% in 4Q20. Earnings results and commentary suggests that fundamentals remain robust across the Software universe, albeit with companies that experienced particularly robust growth during COVID due to macro or product cycle dynamics guiding to a deceleration or normalization in growth rates for 2022 (e.g. SHOP, HUBS, SQSP, AMPL, KLTR). The median ARR growth rate in our sample set was 37.7%, vs. 38.0% in 3Q21 and 34.4% in 4Q20.

Exhibit 1: The median Software company in our sample set beat 4Q21 revenue estimates by 3.0% vs. 3.6% in 3Q21 and 4.3% in 4Q20.
Median beat vs FactSet sales estimates (1Q19-4Q21)



Source: Company data, Goldman Sachs Global Investment Research, FactSet

2) By segment: Security & Workflow automation saw the strongest 2022 estimate revisions, while uCaaS & Productivity were more muted

By category, we note the highest revenue estimate revisions in Security & Workflow automation.

- **Security**: Elevated demand for Security remains more robust than anticipated. As highlighted in our recent mid-quarter recap note, guidance and commentary from Network Security vendors implied more durable growth over the next few years driven by greater volume over traditional networks, share donation from legacy network equipment vendors, product cycles, and new use cases emerging for firewalls. For vendors in Identity and Vulnerability Management, growth is being driven in part by an elevated threat environment, focus on Zero Trust, and greater c-suite/board level awareness of Security posture.
- **Workflow Automation:** This category includes ServiceNow, Monday.com, Unity, Appian and PegaSystems, with outperformance in the quarter driven primarily by Monday.com, Unity and Appian. Workflow automation vendors continue to see strong demand as digital transformation remains a strong tailwind. Larger enterprises in particular continue to focus on automating mission critical workflows by leveraging low-code capabilities. Furthermore, customer expansion continues to be healthy as use cases are expanding for platforms such as Monday.com and

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Appian.

We note the downward revisions in uCaaS, lack of positive revision in Productivity, and more modest revisions in eCommerce. uCaaS 2022 downward revisions were most pronounced at On24, Kaltura and Zoom, largely as a function of navigating more difficult yoy comparisons due to COVID, and the shift in buying patterns from point solutions to solutions likely driving consolidation of spend. In eCommerce, Shopify and BigCommerce revenue estimates were essentially unchanged, as both these companies look to better monetize a larger installed base in the context of difficult 1H22 comparisons.

The Vertical and HCM subsegments saw steady revisions. Many verticals experience longer sales and change management cycles (e.g. insurance, financial services, education), translating to steadier growth for longer. Similarly, in HCM, we believe demand tends to be steadier and more likely to be tied to overall macro trends on hiring than investment priorities in any given year.

In CRM, many companies commented that they did not believe there was a pull forward in demand in 2021, and that the demand environment remains healthy — with 2022 estimates revised up 2.8% at the median. These comments are consistent with our own industry conversations and our December 2021 CRM budget survey. More broadly, we believe companies continue to invest in their CRM technology stacks, with incremental growth driven by a) cross CRM functionality, to aggregate data across disparate sales, marketing, and service silos; b) omni-channel engagement, with in person returning as a channel to complement web, phone, text and app engagement; c) the ROI offered by next-generation tools that build on top of Salesforce and Adobe deployments, such as Qualtrics (for customer experience), Braze (for marketing), and select private companies with outsized momentum (e.g. Klaviyo, Gong); d) the shift from third party data to first party data requiring incremental investment in data ingestion, personalization and classification.

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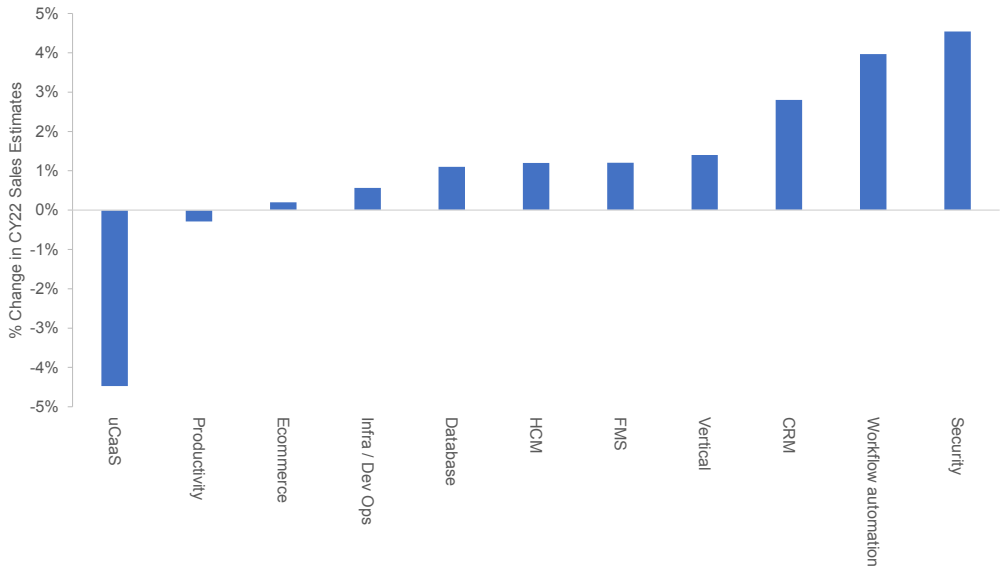
Exhibit 2: The Security & Workflow Automation categories are seeing the strongest revisions
Change in 2022 sales estimates (FactSet), median by category, 12/14/21 to 3/2/22



Revenue revision normalized for M&A

Source: FactSet, Data compiled by Goldman Sachs Global Investment Research

3) 4Q marks a reversal in CAC trends

Gross cost of customer acquisition (CAC) increased slightly to 1.23 in 4Q from 1.20 in 3Q, marking a reversal in trend from 5 quarters of declines averaging 4% qoq. The increase was largely driven by an increase in TTM S&M expense growth, at 43% yoy vs. 25% in 4Q20. We expect CAC to trend higher in 2022, given a normalization in demand for companies that benefited from COVID tailwinds, a return to more in person events such as sales kick-offs and customer conferences, investments in hiring (after likely delays / pushouts in 2020 and 2021), and increased wage inflation.

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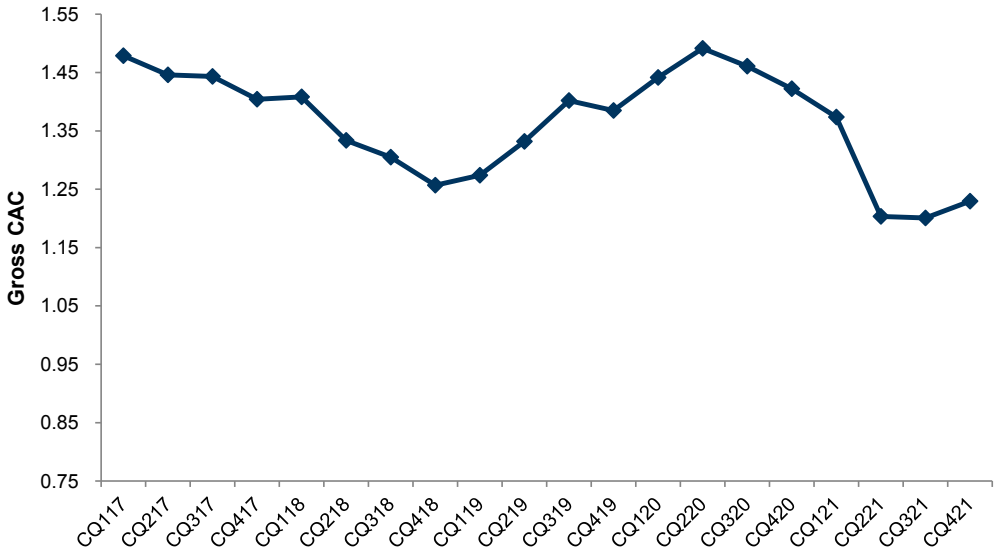
Exhibit 3: Gross CAC continued to decline in C4Q
Average Gross CAC



Includes all tickers listed in Exhibit 10 except outliers INTU/SWI; CRWD, S, YOU, AVPT data is as of 3Q21

Source: Company data, compiled by Goldman Sachs Global Investment Research

4) Hiring and compensation costs are increasingly top of mind

Companies acknowledge a tough environment, but few specifics on hiring and retention strategies. Several companies acknowledged the challenges with hiring into 2022, and we believe the bid for quality talent both in the US and at international technology hubs has taken a step up from already elevated levels, in part due to increased flexibility around remote work. The challenge may be further compounded by recent volatility in stock prices as this could lead to increased turnover ahead of typical stock vesting cliffs (as the stock "left on the table" is now worth less). This is consistent with the Wall Street Journal reporting higher cash packages being offered by tech companies to offset volatility in stock based comp (link, 2/27). Companies generally spoke to their unique cultures and competitive compensation packages as allowing them to continue to execute on hiring plans, but gave few specifics (Exhibit below). We note that Workday quantified a 300 bps headwind to EBIT margin in 2022 due to increased compensation costs. We believe the most challenging set ups will be those companies that are hiring into a product cycle slow down, as increased sales count may not necessarily address a demand-driven issue.

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Exhibit 4: Hiring and compensation are top of mind, with companies highlighting culture and incremental changes to compensation plans

Select comments from 4Q earnings calls, abridged, please see transcripts for full details

Company	Comments
Avalara	• Hiring is harder due to higher pay and return to office • Have not seen a change in retention rates, employee engagement is the same as before the pandemic, maybe better • Have had success in finding people, added 1,500 people last year • Not as many challenges in S&M hiring as in hiring developers, which is always difficult • Have found ways to keep people and to add the people needed to grow
Hubspot	• Talent and especially SaaS talent has been in high demand for a while, hard to remember when this was not the case • Hubspot has foundational factors allowing them to hire and retain talent even in this market- culture and providing flexibility • Provided flexibility and made hybrid work for teams, this has been key for them
ServiceNow	• Leadership team continues to strengthen, loyal global workforce with high retention rates • Recruiting great talent and keeping it, much lower attrition in engineering than peer group benchmarks • No company is immune to war on talent, culture differentiates them • Hire rankings on Glassdoor show how invested the company's people are in the future of ServiceNow
Shopify	• Tech talent is in high demand, desire to remain competitive • SBC outlook reflects full year impact of hiring last year, hiring they will do this year, and expected adjustments • Did more than double R&D hiring last year yoy which indicates that Shopify remains the best place in tech to work due to compensation, the exciting things the company is doing, and continued innovation
Workday	• Implemented company-wide cash bonus program, historically only in sales and services organizations and compensated through being generous with stock • Stock only is not enough now, always planned on eventually layering in bonus program across the company and have decided to do so now • Impact will be in R&D which is about half the company, as well as G&A • The cost of this is 300 bp • Important program, excited for employees and believe it will help with retention and recruiting

Source: Company data, FactSet, Data compiled by Goldman Sachs Global Investment Research

5) Rates and Multiple Compression

Multiples back to 2019 levels, but stronger business models today arguably warrant a premium. The average EV/sales/growth multiple for our sample set is 0.38x (NTM), slightly above median levels from 2H19. 2H19 was the last time 10-year rates were in the 2.0-2.5% range (which would be consistent with GS expectations for 2023 and 2024). Relative to 2019, we believe the Software sector is stronger today, given a) a larger software TAM — e.g, we estimate the cloud TAM for front office is $72 bn in 2022 vs. $40 bn in 2019; b) more evidence of durable growth at scale — the average company in our sample set is growing from a revenue base that is twice as big as our sample set in 2019; c) increased evidence of growth durability through macro shocks given fundamental performance through 2020 and 2021; d) improved efficiency, with LTV:CAC for the group at 4.06x vs. 3.54x in 2019 (albeit with some reversion back to trend in 2022 given the margin and cost dynamics we discuss above).

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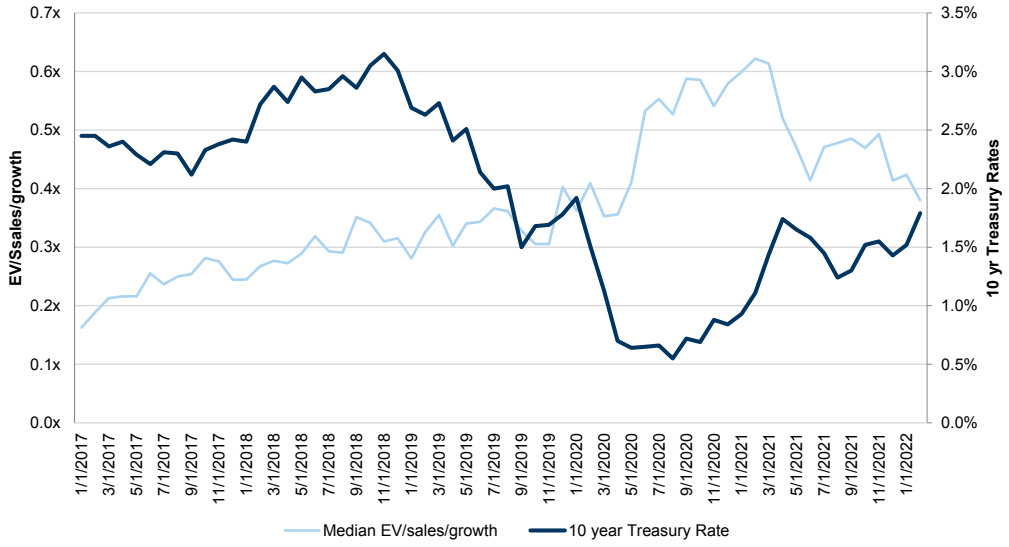
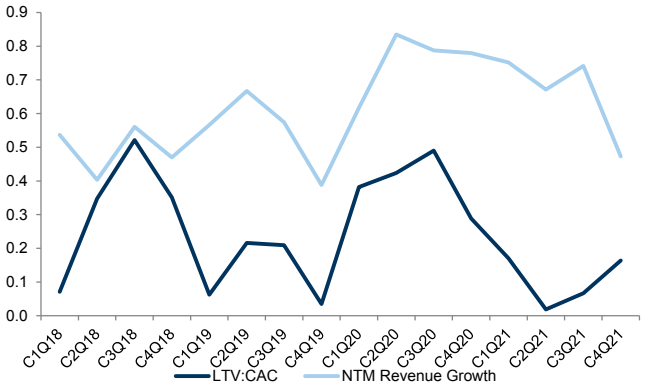
Exhibit 5: Top quartile EV/sales/growth multiples for our sample set are essentially back at 2H19 levels - the last time rates were 2.0-2.5%
EV/sales/growth vs 10 year treasury rates



Source: FactSet

Valuation correlations with LTV:CAC improved, and worsened vs. revenue growth. The correlation of revenue growth to EV/sales multiples decreased in 4Q21 vs. 3Q21, while the correlation of LTV:CAC to EV/sales multiples increased. Through most of 2020 and 1H21, we believe investors were rewarding revenue growth as the secular demand for digital transformation was strong relative to more mixed macro trends. In contrast, today, we believe investors are more cognizant of the quality of growth (and whether it can be sustained post COVID-catalyzed product cycles), and the efficiency of the business and FCF.

Exhibit 6: Correlation to unit economics increased in C4Q21 while correlation to revenue growth declined
Correlation of EV/Sales with unit economics and NTM revenue growth



Source: Company data, FactSet, Data compiled by Goldman Sachs Global Investment Research

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Benchmarking KPIs

For the broader software sector, we believe it is important for high-growth (and high-multiple) companies to show potential for meaningful profitability at scale. Therefore, we track and rank order the unit economics of our coverage to get a better sense of potential profitability and what valuation is warranted relative to peers.

- **Lifetime Value (LTV).** Our LTV equation aims to derive the discounted value of $1 in recurring revenue by taking into account customer churn (we used GS estimates where not disclosed) and subscription gross margins (the cost to service $1 of recurring revenue). We recognize the ongoing costs related to servicing existing customers may be materially lower than reported gross margins; however, we have limited insight into gross margin by customer cohort across our coverage universe. We also utilize a discount rate in our equation to reflect the longer-term displacement and disruption risk inherent in software.
- **Customer Acquisition Cost (CAC).** Our CAC equation estimates the S&M expense required to generate a new dollar of annualized recurring revenue (ARR), which we view as a proxy for bookings. For costs, we use GAAP sales and marketing, as we believe stock-based compensation is a requisite cost of doing business in the software industry. We do not adjust for deferred commissions, as we believe the penalties/rewards of deferred commissions normalize over time. For companies in which ARR is not disclosed, we annualize quarterly subscription revenue (with a few exceptions) and take the y/y change in ARR. We also add back gross churn to calculate the gross new ARR prior to churn. We recognize that gross new ARR, as we measure it, is not a perfect leading indicator, but other preferred metrics such as new current billings are impacted by variable contract terms. Internal calculations of CAC may be on a per customer or ACV basis, but with the available disclosures we view this as the best method of comparing CAC across our coverage universe.

Exhibit 7: Our methodology
LTV: CAC formulas

$$LTV = \frac{GAAP\ subscription\ gross\ margin}{Gross\ churn\ (est.) + 8.5\%\ discount\ rate}$$

$$CAC = \frac{TTM\ GAAP\ sales\ \&\ marketing\ expense}{\frac{y}{y}\ gross\ new\ ARR}$$

Source: Goldman Sachs Global Investment Research

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Exhibit 8: 4Q21 benchmarking software: Summary table

Sorted from highest-to-lowest EV/ARR; ARR = Annualized Recurring Revenue (as at end 4Q21); EV and ARR in $ mn

	Price	EV	EV/Sales CY22	Gross Churn	Discount Rate	Gross CAC	LTV	LTV: CAC	Latest NRR	ARR	EV/ARR	ARR Growth
SNOW	$209.65	$72,148	37.0x	3%	9%	$1.00	6.9x	6.85	178%	$1,438	50.2x	102%
S	$36.74	$9,685	29.0x	3%	9%	$1.03	5.5x	5.35	130%	$237	40.9x	131%
DDOG	$142.19	$48,334	31.8x	7%	9%	$0.46	5.1x	11.02	NA	$1,305	37.0x	84%
ZS	$225.50	$33,102	27.5x	6%	9%	$1.35	5.3x	3.97	125%	$1,022	32.4x	63%
CRWD	$179.03	$42,181	21.6x	3%	9%	$0.89	6.6x	7.41	125%	$1,514	27.9x	67%
TEAM	$267.61	$69,585	23.0x	14%	9%	$0.46	3.8x	8.34	NA	$2,544	27.4x	44%
ZI	$53.00	$22,768	22.2x	10%	9%	$0.62	4.4x	7.16	116%	$892	25.5x	60%
U	$89.01	$29,836	20.0x	5%	9%	$0.83	5.7x	6.81	140%	$1,178	25.3x	46%
CFLT	$35.60	$9,305	17.0x	10%	9%	$1.56	3.8x	2.43	130%	$433	21.5x	71%
NOW	$547.07	$111,392	15.0x	3%	9%	$1.56	7.5x	4.85	NA	$6,092	18.3x	29%
OKTA	$158.41	$25,964	14.9x	4%	9%	$1.26	5.8x	4.61	124%	$1,477	17.6x	64%
PANW	$557.51	$65,938	11.1x	5%	9%	$1.73	5.1x	2.97	NA	$4,036	16.3x	32%
MNDY	$135.91	$5,965	12.6x	10%	9%	$1.33	4.8x	3.61	120%	$382	15.6x	91%
HUBS	$459.03	$22,237	12.9x	13%	9%	$1.12	4.0x	3.57	NA	$1,435	15.5x	47%
CYBR	$160.24	$6,068	10.3x	5%	9%	$2.07	6.1x	2.97	NA	$393	15.4x	43%
CWAN	$17.50	$4,286	14.2x	2%	9%	$0.63	6.8x	10.79	111%	$278	15.4x	26%
XM	$27.50	$14,676	10.4x	10%	9%	$1.40	4.6x	3.30	NA	$1,036	14.2x	61%
PCOR	$60.75	$8,179	12.3x	6%	9%	$1.79	5.5x	3.09	107%	$584	14.0x	33%
IOT	$15.52	$7,764	13.8x	10%	9%	$1.16	3.8x	3.26	115%	$558	13.9x	64%
DH	$20.34	$2,544	11.6x	14%	9%	$0.82	3.4x	4.15	111%	$183	13.9x	38%
INTU	$464.95	$141,146	10.8x	13%	9%	$0.62	3.9x	6.30	NA	$10,692	13.2x	70%
BSY	$38.04	$12,161	10.9x	2%	9%	$0.88	7.4x	8.38	109%	$921	13.2x	22%
DT	$41.43	$12,080	11.0x	8%	9%	$1.28	5.5x	4.27	123%	$930	13.0x	29%
PLAN	$49.09	$7,591	10.4x	7%	9%	$2.20	5.4x	2.45	118%	$592	12.8x	32%
WDAY	$240.21	$62,227	10.2x	4%	9%	$1.42	6.6x	4.69	NA	$4,917	12.7x	22%
SHOP	$600.84	$69,622	11.5x	16%	9%	$0.40	2.1x	5.11	NA	$5,520	12.6x	41%
RPD	$103.80	$7,545	11.0x	6%	9%	$1.29	4.9x	3.78	112%	$599	12.6x	38%
AVLR	$94.89	$8,130	9.5x	5%	9%	$1.69	5.3x	3.16	116%	$679	12.0x	28%
YOU	$22.08	$3,223	8.5x	15%	9%	$0.39	3.6x	9.29	NA	$270	11.9x	20%
SAIL	$45.25	$4,389	8.5x	5%	9%	$1.79	5.7x	3.19	NA	$370	11.9x	47%
TENB	$55.57	$6,589	9.9x	5%	9%	$1.86	5.9x	3.15	110%	$566	11.6x	28%
PYCR	$27.24	$4,738	10.6x	9%	9%	$1.43	3.4x	2.41	88%	$411	11.5x	20%
KNBE	$19.17	$3,199	9.7x	11%	9%	$0.99	4.4x	4.48	108%	$285	11.2x	44%
QLYS	$128.26	$4,833	10.0x	5%	9%	$0.97	5.8x	5.99	NA	$439	11.0x	16%
JAMF	$32.72	$4,446	9.5x	5%	9%	$1.05	6.0x	5.69	120%	$413	10.8x	45%
ESTC	$79.65	$7,728	7.7x	13%	9%	$1.15	3.7x	3.27	129%	$838	9.2x	42%
PWSC	$15.77	$4,361	7.0x	5%	9%		5.2x	NA	106%	$539	8.1x	26%
CRM	$203.01	$207,351	6.6x	9%	9%	$1.63	4.7x	2.86	NA	$27,312	7.6x	25%
SPLK	$129.06	$23,646	7.4x	6%	9%	$1.72	5.5x	3.18	132%	$3,117	7.6x	32%
ZM	$108.94	$28,270	6.3x	23%	9%	$0.73	2.5x	3.37	NA	$4,286	6.6x	21%
PING	$20.09	$1,876	5.6x	8%	9%	$1.58	5.0x	3.15	112%	$313	6.0x	21%
BIGC	$20.94	$1,588	5.7x	8%	9%	$0.98	4.7x	4.82	NA	$269	5.9x	48%
AVPT	$5.07	$788	3.1x	7%	9%	$2.29	5.6x	2.46	110%	$148	5.3x	32%
SWI	$12.34	$3,239	4.4x	8%	9%		4.4x	NA	104%	$631	5.1x	NA
INFA	$17.65	$6,295	3.9x	10%	9%	$1.57	4.2x	2.67	114%	$1,360	4.6x	17%
EVBG	$36.46	$1,671	3.9x	13%	9%	$1.23	3.4x	2.75	110%	$361	4.6x	37%
DBX	$22.00	$8,357	3.6x	17%	9%	$0.74	3.2x	4.32	NA	$2,262	3.7x	12%
KLTR	$2.05	$156	0.9x	10%	9%	$0.92	4.0x	4.27	120%	$154	1.0x	33%
Average			**12.2x**								**14.9x**	**43%**
Median			**10.5x**								**12.7x**	**38%**

Prices as of 3/7/22; CRWD, S, YOU, AVPT data is as of 3Q21; PWSC excluded given sales expense reported within SG&A

Source: Company data, FactSet, Goldman Sachs Global Investment Research

Company-specific trends

We highlight a few select names that stand out to us with improving or increasing CAC.

Most notable decreases in CAC:

- **XM:** Qualtrics' CAC benefited from its acquisition of Clarabridge at the start of 4Q, as it was down 0.67x yoy. Adjusting for M&A, we believe the company still screens

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well for organic improvement in CAC, at down 0.41x yoy, and we see a scenario where Qualtrics is able to maintain or improve its organic CAC over the course of 2022 (vs. initial guidance implying a decline). We believe improving CAC is a function of momentum in customer experience investments, coupled with improving productivity for enterprise and international reps.

- **PYCR:** We estimate CAC decreased by 0.57x yoy and by 0.10x qoq. Paycor's CAC is improving due to ramping productivity of sales hires made over the last several quarters and improving traction in Tier 1 regions, which is driving accelerating bookings and revenue growth.
- **TENB & RPD:** Both Vulnerability Management vendors are having success expanding into adjacent markets while increasing channel engagement, improving penetration up market and driving greater revenue per customer in the process.
- **MNDY:** While we estimate Monday's CAC decreased by 0.50x yoy and 0.06x qoq as a result of the company's revenue outperformance, the company's near-term focus on new products and its go-to-market strategy may result in an increase of CAC until such efforts ramp. We forecast the company's sales efficiency, measured as S&M expense divided by incremental revenue, coming back up to the ~2.0x range in 2022 after this metric averaged ~1.5x from 2Q21-4Q21.

Most notable increases in CAC:

- **ZM:** We estimate that Zoom Video's CAC increased by 0.50x yoy and by 0.16x qoq, primarily as a result of increase in sales & marketing programs and sales hiring driving up costs. ZM was a huge COVID beneficiary, now as sales revenues decelerate on the back of tough comps and COVID tailwinds wane, the company is capitalizing on the brand momentum gained during the pandemic by investing aggressively in S&M, particularly on the international side and Zoom Phones.
- **AVLR:** We estimate that Avalara's CAC, when adjusted for 2021 M&A, increased by 0.33x yoy and by 0.06x qoq, largely due increased investment in go to market and product in 2021 and into 2022. (CAC is about flat yoy without making an adjustment for M&A). We see a path to stronger CAC over time from bundling, cross sell, and expansion in the enterprise, as 2021 and 2022 investments become productive.
- **SHOP:** We estimate that Shopify's CAC increased by 0.13x yoy while remaining flat qoq. We view this as a function of normalizing demand post COVID-catalyzed ecommerce adoption. We believe the key questions going forward are a) whether Shopify can maintain unit economics comparable or better than pre-COVID levels, b) ROI associated with ongoing fulfillment expenses.
- **PCOR:** We estimate Procore's CAC increased by 0.20x yoy driven by the reacceleration of sales and marketing in the quarter and the 47% increase in sales and marketing headcount this year. As these investments ramp and growth normalizes vs. lighter investments in 2020, we see this metric improving. CAC's qoq decline of 0.11x qoq is in line with this thesis.

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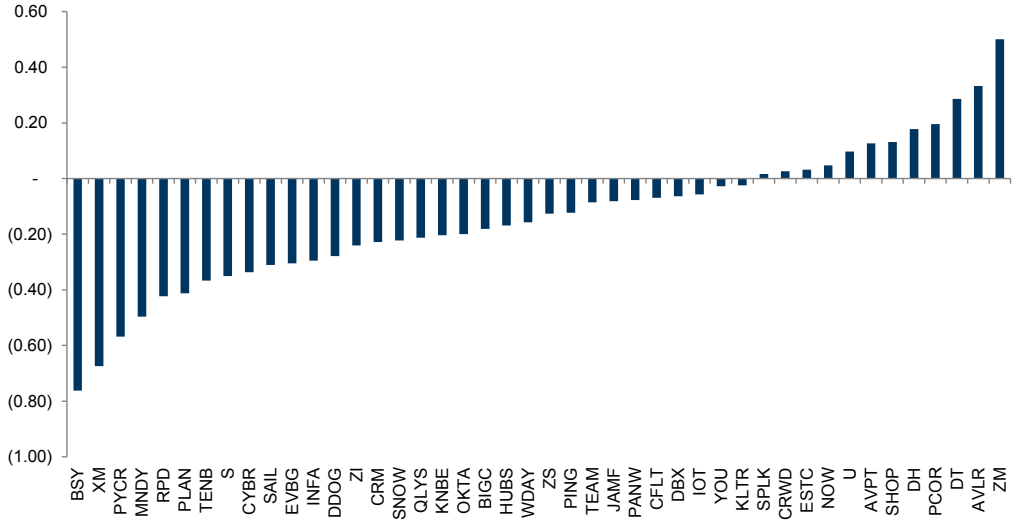
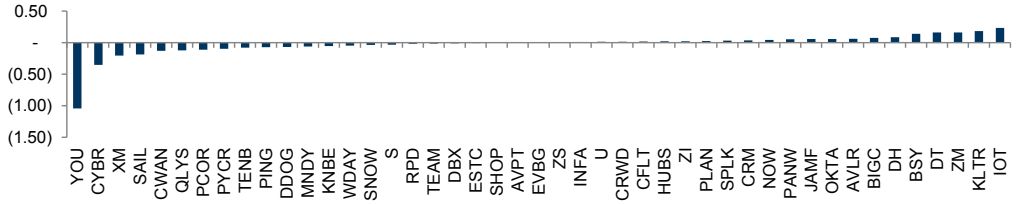
Exhibit 9: CAC change 4Q21 – y/y
(improving) / deteriorating



Excluding INTU, SWI; CRWD, S, YOU, AVPT data is as of 3Q21

Source: Company data, compiled by Goldman Sachs Global Investment Research

Exhibit 10: CAC change 4Q21 – q/q
(improving) / deteriorating



Excluding INTU, SWI; CRWD, S, YOU, AVPT data is as of 3Q21

Source: Company data, compiled by Goldman Sachs Global Investment Research

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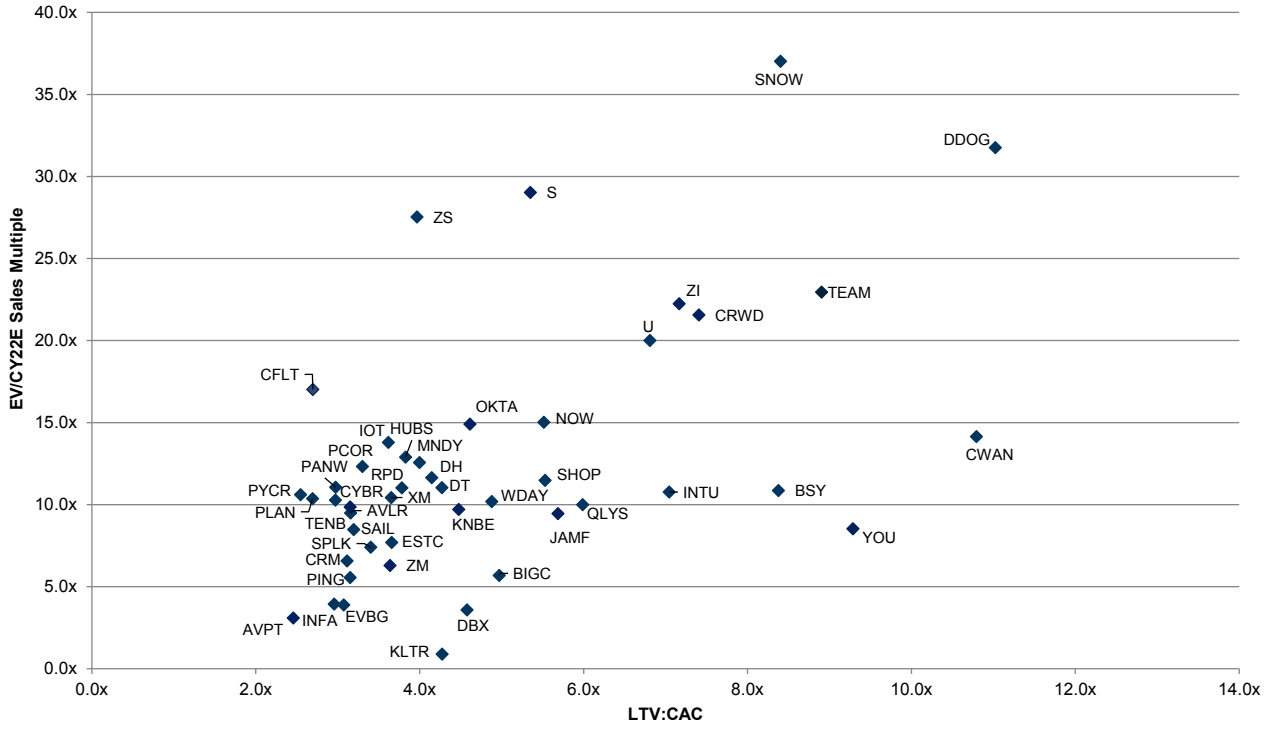
Correlating unit economics and efficiency

Exhibit 11: LTV:CAC vs. EV/CY22E Sales
Priced as of 3/7/22



S,CRWD, YOU, AVPT LTV:CAC as of 3Q21, Excluding SWI, PWSC

Source: Company data, FactSet, Goldman Sachs Global Investment Research

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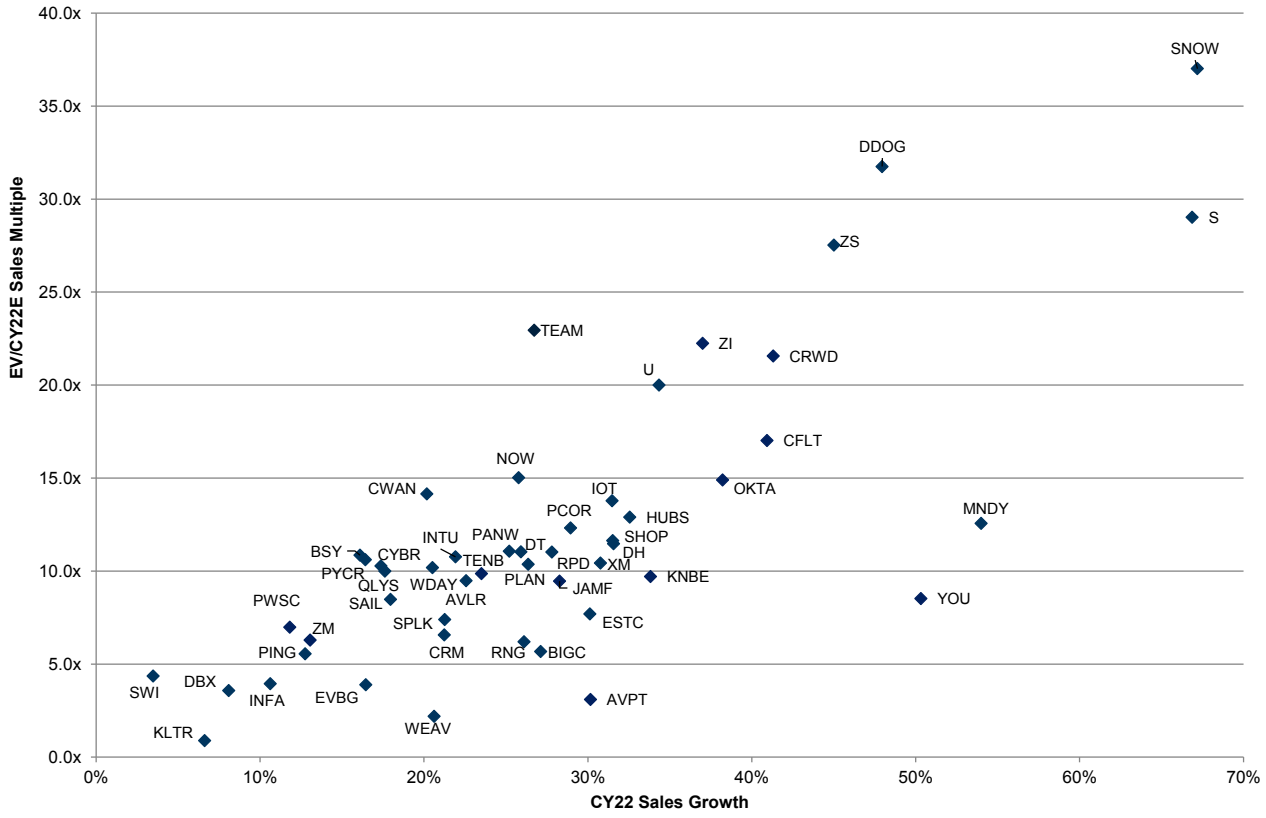
Exhibit 12: CY22E Sales Growth vs. EV/CY22E Sales
Priced as of 3/7/22



Source: Company data, FactSet, Goldman Sachs Global Investment Research

Rule of 40

Our analysis of the "Rule of 40" suggests an increasing degree of correlation between CY22E EV/Sales and the Rule of 40 (R^2 of 0.31 vs 0.21 in 3Q). For reference, the Rule of 40 is typically defined as the sum of revenue growth and FCF margins, and can be used as a proxy for "efficient growth." Companies with best-in-class unit economics typically screen favorably on the Rule of 40 since they tend to grow more efficiently than peers. Some companies that screen well on the Rule of 40 include SNOW, ZI, CRWD, ZS and DDOG.

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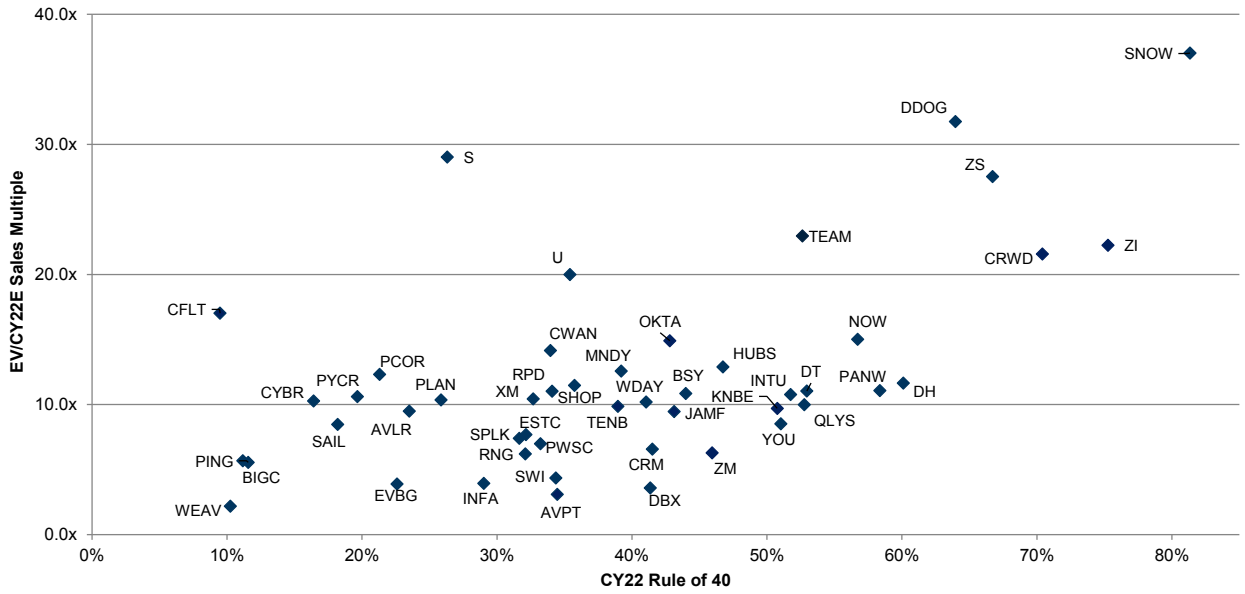
Exhibit 13: Rule of 40 (CY22E Revenue Growth + FCF) vs. EV/CY22E Sales
Priced as of 3/7/22



Source: Company data, FactSet, Goldman Sachs Global Investment Research

Methodology

We view subscription software models as net present value equations. Simplistically, the sales and marketing costs to acquire a dollar of recurring revenue (CAC) are the initial investment, and the recurring revenue streams excluding support costs (COGS, R&D, and G&A) are the returns. We view CAC as a particularly important metric for two reasons: 1) CAC tends to be predictive of longer-term cash flow generating potential, all else equal, and 2) CAC tends to trend higher over time as the companies' core markets mature and they expand into new geographies and adjacencies as a result. Starting with a lower initial CAC provides more headroom for sustained investment without sacrificing cash flow generation as CAC increases.

Our methodology is as follows.

1. *For our costs, we use GAAP sales and marketing.* We view stock based compensation as a cost of doing business, and for most of our companies under coverage, stock compensation is the primary delta between GAAP and non-GAAP expenses. We do not adjust for deferred commissions, which can penalize some models and reward others (e.g. NOW). Over time, we think this normalizes.
2. *For a new annualized contract value (ACV) bookings proxy, we use gross new ARR.* To arrive at new ARR, we annualize the year-over-year dollar change in recurring revenue for the most recent quarter. We define recurring revenue as subscription revenue, with a few exceptions. For SHOP, we include merchant solutions revenue. New ARR as we measure it is a somewhat lagging measure, and we view new

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current billings as a more accurate measure. However, as many subscription companies have variable billings terms, using new current billings is challenging. Generally, current billings and ARR track each other closely. We also adjust for churn.

3. *We do not adjust for currency, as disclosure of top-line versus bottom-line headwinds is inconsistent in software.*
4. *We do not adjust for M&A.*

LTV to CAC

LTV to CAC is a popular measure of unit economics in SaaS. There are many different ways to define and calculate LTV to CAC, with many of them derived on a per customer basis or on an annualized contract value (ACV) basis. However, many companies in our cohort do not disclose customer count or customer retention on a regular basis. ACV is also not typically disclosed. Therefore, our definitions and calculations of LTV and CAC are as follows.

- LTV: gross margin / (gross revenue churn + discount rate of 8.5%). Gross revenue churn is calculated as 1- revenue retention. Our aim when calculating LTV is to derive the discounted value of $1 of recurring revenue. The two most integral inputs into our LTV equation are churn (which reduces the future value of $1 each year) and gross margins (which reflect the cost to service the $1 of recurring revenue). We also use a discount rate in our equation to reflect the longer-term displacement and disruption risk inherent in software.
- Gross CAC: TTM GAAP S&M expense / Gross new ARR. Gross new ARR is calculated as current quarter annualized recurring revenue – year ago quarter annualized recurring revenue. We add back gross ARR churn back into new ARR based on our assumption for gross churn. This CAC ratio measures how much in S&M expense is required to generate a new dollar of gross ARR, prior to churn, which we view as a proxy for gross bookings.

We interpret a higher LTV:CAC ratio to mean that a company's "growth equation" is more value generative. A lower CAC (and associated higher LTV:CAC ratio) generates more cash flow, faster when all other inputs are held constant.

Since not all companies disclose retention – and for those that do, their definition of retention varies – we have estimated a range of gross revenue retention for our analysis. Our estimates are based on comparables, end markets, and, in some cases, company commentary at conferences or on earnings calls. We excluded from our LTV to CAC analysis the companies that we believe have not provided enough information on revenue retention.

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We, Gabriela Borges, CFA, Kash Rangan, Brian Essex, CFA, Nishtha Poddar, Kevin Kumar, CFA, Carolyn Valenti, Jake Titleman, Anisha Narayan, CFA, Gili Naftalovich, Charlotte Bedick and Hannah Velasquez, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

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Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

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